

RESEARCH ARTICLE

Programming Interfacial Polymerization: Machine Learning Unveils Quantitative Rational Design Rules for Microcapsules and Beyond

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Received: 9 September 2025 | **Revised:** 22 December 2025 | **Accepted:** 5 January 2026

Keywords: interfacial polymerization | interpretable machine learning | microencapsulation | quantitative chemical-process-structure-performance | rational design descriptor

ABSTRACT

Interfacial polymerization (IP) serves as a versatile platform technology for designing polymeric membranes, yet its extension to applications such as microencapsulation (MIP) remains hindered by empirical methodologies, largely due to the absence of quantitative rational design principles. Unlike separation membranes, which prioritize nanostructural control, MIP emphasizes encapsulation efficiency (EE%), rendering conventional membrane-derived theories and thermodynamic descriptors insufficient. In this work, we transcend these limitations by employing interpretable machine learning to program interfacial polymerization, thereby deciphering mechanism-informed quantitative design rules. Our data-driven platform integrates molecular thermodynamics, polymerization kinetics, and emulsion-stabilized interfacial parameters to identify previously overlooked descriptors governing microcapsule formation. We establish a predictive chemical–process–structure–performance relationship and demonstrate programmable control over key performances, including EE% (30%–95%), particle size (100–400 μm), and shell thickness-to-radius ratios (0.005–1) for diverse payloads spanning hydrophobic, hydrophilic, and highly reactive compounds such as toluene diisocyanate and amines. This work not only resolves long-standing challenges in understanding complex multiphase interactions in MIP but also establishes a new paradigm for the quantitative design of polymeric microcapsules, with broad implications for functional particles, catalytic microreactors, digital cells, and membranes.

1 | Introduction

Interfacial polymerization (IP), a reaction confined to liquid–solid, liquid–liquid, or liquid–gas interfaces, has become a plat-

form technology of advanced membrane materials innovation [1, 2]. Its applications span a wide range of synthesis timescales and complexities, from ultrafast advanced membrane formation (minutes) for selective separation [3–5], to hours-long

microencapsulation processes in emulsions for multifunctional microcapsules [6–9]. Despite its versatility, the field faces a stark dichotomy. Short-timescale IP for separation membranes has seen significant progress in mechanistic and systematic studies [10–14], while longer-timescale microencapsulation by interfacial polymerization (MIP) remains unclear. MIP encompasses intricate couplings among multicomponent thermodynamic interactions, reaction-diffusion competition-induced shell polymerization kinetics, polymerization-induced phase separation, and emulsion droplet hydrodynamics. Despite its broad potential, the development of IP for new applications is fundamentally constrained by a reliance on empirical approaches. A critical, unresolved challenge is the lack of quantitative rational design rules that can universally link critical mechanisms [15], synthesis parameters to structural outcomes and functional performance across the entire IP platform.

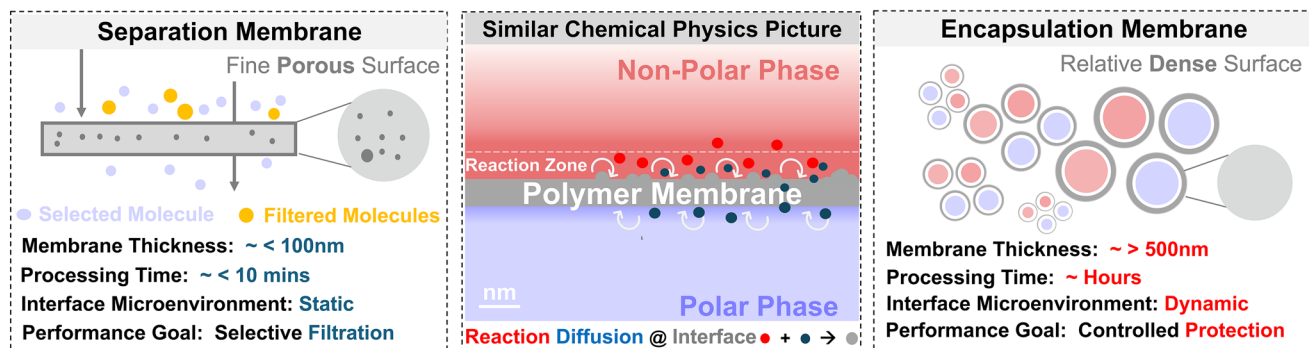
Current understanding of IP predominantly stems from thin-film composite membrane studies, where reactions are completed within a few minutes under static conditions. These works, focusing on kinetic investigation [16, 17] and thermal dynamics regulation [17–19], cannot be directly applied to MIP due to the unique complexities of emulsion media [20] including: 1) competing absorption, reaction and diffusion of molecules at oil-water interface [21], 2) heterogeneous microenvironment affected by process conditions and hydrodynamics induced droplet deformation introduces localized variations of molecules and energy transport [22]. A key divergence also lies in the performance objectives. Separation membranes are designed and optimized for precise selective filtration and purification of gases or small molecules, relying on tightly controlled nanostructures [3–5, 10]. Encapsulation membranes, by contrast, aim to maximize the encapsulation of target molecules, prioritizing high encapsulation efficiency and relatively dense nanostructure for controlled protection [23–25]. These fundamental differences in processing timescale, the complexity of physical and chemical microenvironments [19, 20, 22], and performance goals render existing separation membrane theories and tools inadequate for designing MIP-based encapsulation systems, as shown in Figure 1a. Consequently, microencapsulation by interfacial polymerization processes often relies on empirical recipes rather than predictive design (Figure 1b). Although numerous quantitative studies have been conducted to predict shell thickness [26–28] and design the microstructures [29] of microparticles synthesized via interfacial polymerization, their applicability remains constrained by a limited molecular design space and insufficient emphasis on encapsulation efficiency (EE%). In this study, EE% was selected as the primary criterion for model development and formulation screening [25, 30]. Encapsulation efficiency is directly tied to the core content of microparticles and the yield of the microencapsulation process, and it serves as the foundation for achieving controlled release, functionalization in specific application scenarios, and material stability under harsh or long-term storage conditions. Therefore, assessing EE% is essential for the effective design of encapsulation membrane systems, as it provides critical insight into the mechanisms governing shell formation, microstructural evolution, and the development of new multifunctional capsules. For the targeted application scenarios in this work (e.g., delivery of active ingredients, functional additives in coatings, and storage of reactive cores), EE% directly

determines the usable payload per unit mass of microcapsules, the effective dosage or activity in the final formulation, and the extent of core loss into the surrounding medium. By contrast, μD (average diameter of microparticles) and h/R (shell radius ratio of core-shell type microparticles) mainly describe particle dispersion and shell geometry, which are crucial for processability and release behavior but can often be tuned in subsequent steps (e.g., by adjusting shear conditions, surfactant content, or post-processing) once a sufficiently high EE% has been achieved. Consequently, we first prioritize EE%-based descriptor discovery and then use the μD and h/R models as complementary tools to refine particle size and shell architecture within the high-EE% formulation window.

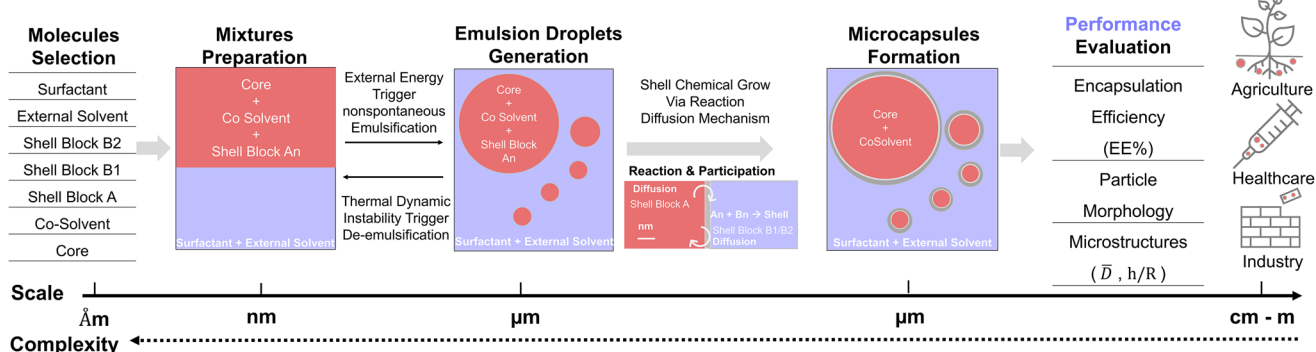
The design of encapsulation membrane-based microparticles has traditionally been guided by thermodynamic-focused descriptors, such as Hildebrand solubility parameters [31], Hansen solubility parameters [32], Flory-Huggins interaction parameters [33], viscosity, refractive index, and interfacial tension [34]. While these descriptors establish correlations between molecular properties and encapsulation efficiency, they have notable limitations: (1) incomplete molecular design space: Most of these parameters describe thermodynamic interactions rather than intrinsic molecular attributes, requiring computationally expensive combinatorial calculation and diverse, labor-intensive experimental characterization. This complexity limits the exploration of molecular systems, leaving much of the design space untapped; (2) neglect of microenvironmental and reaction kinetic factors: Insufficient attention has been given to the role of surfactant-assisted emulsion stability, polymerization effects and hydrodynamic effects in influencing encapsulation performance; (3) complex interdependencies: The intricate interactions among chemical properties, process parameters, microstructures, and performance metrics remain poorly understood, particularly for MIP systems. Addressing these challenges requires robust methodologies that can navigate the multifaceted complexities of encapsulation membrane material systems characterized by multiple coupling factors. Machine learning has proven to be a transformative tool for addressing such material design challenges, offering capabilities for systematic novel matter discovery [35], predictive design [36], hidden knowledge extraction [37, 38], and structure optimization [39].

To overcome these limitations, this study proposes a data-driven framework that uses interpretable machine learning to correlate molecular thermodynamics (e.g., solubility and partitioning), polymerization-kinetics descriptors, and emulsion-based interfacial-polymerization parameters. This approach establishes a quantitative chemical–process–structure–performance (CPSP) linkage, elucidating the complex mechanisms of MIP and enabling high-throughput, rational design of microcapsules (Figure 1b). The framework is validated by extrapolating to encapsulation systems for hydrophobic and hydrophilic molecules, as well as highly reactive species such as 2,4-toluene diisocyanate (TDI) and amines. It further supports multi-output performance prediction, including encapsulation efficiency, particle size, membrane shell thickness, and the morphological state of the particle. Beyond improving microcapsule design, the methodology also provides mechanistic feedback that can inform the development of separation membranes formed by interfacial polymerization.

a). Interfacial Polymerization Based Membrane Systems



b). Microencapsulation by Interfacial Polymerization To Form Encapsulation Membrane System



c). Data Driven Microencapsulation by Interfacial Polymerization

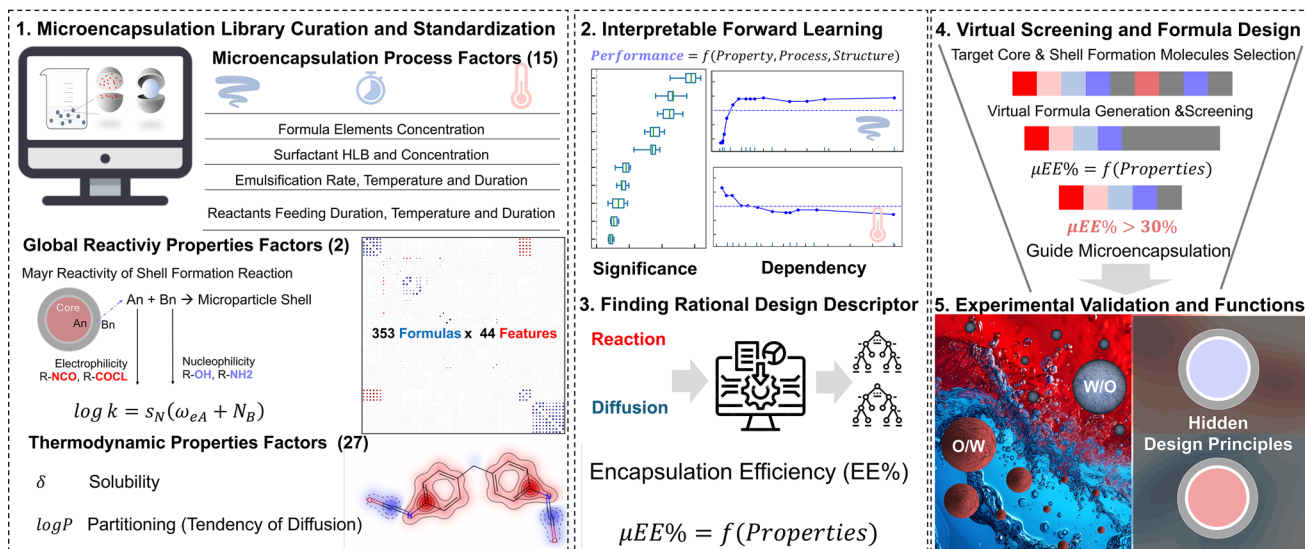


FIGURE 1 | A physics informed, data-driven platform for predictive microencapsulation design. (a) The main differences between separation and encapsulation membranes made by interfacial polymerization. (b) The physical scale and complexity illustration of the conventional microencapsulation by interfacial polymerization. (c) The informatization process of microencapsulation by interfacial polymerization.

2 | Results and Discussion

2.1 | Feature Standardization and Database Curation

Microcapsule synthesis via interfacial polymerization is a complex, multi-scale, and multi-composition process involving a series of chemical and physical events at a heterogeneous interface (Figure 1a,b). In most laboratory-scale MIP experiments, the process typically involves the following steps: (i) selection and

design of the molecular components of the formula; (ii) preparation of distinct liquid phases: the continuous phase (a mixture of solvent and surfactants) and the dispersed phase (comprising the target core molecule, a core co-solvent, and a portion of the shell-forming material) [15, 40]; (iii) formation of emulsion droplets, using external energy sources such as ultrasonic homogenization or mechanical stirring to create a shear field [41, 42]; (iv) initiation of polymeric shell formation around the droplets by introducing an additional shell-forming material, which undergoes reaction and diffusion at the nanoscale [1, 3, 15, 25, 26]; (v) polymerization

and thermally induced phase separation and solidification of the polymeric shell [27, 28, 43]; (vi) evaluation and characterization of the resulting microcapsules to assess batch performance and their properties; (vii) exploration of the potential applications of the tailor-made MIP products in various fields.

The development of microencapsulation processes is influenced by various experimental factors, including laboratory-scale setup and molecular properties. However, a tailored experimental database for polymeric shell microcapsules made by MIP is currently lacking. To address this knowledge gap, we compiled a systematically curated and comparatively large database by extracting and reconstructing relevant data from the existing literature on MIP-based fabrication of single-layer polymeric shell microparticles. This database enables correlation of experimental conditions and molecular characteristics with process outcomes, providing valuable insights into microencapsulation mechanisms. According to the diversified research interest among different researchers, there are five criterion for constructing the MIP database: (1) at least providing the information of core fraction towards different batches of the experiment; (2) providing the encapsulation efficiency towards different batches; (3) providing the structural information about microparticle size distribution and cross-section of individual particles; (4) providing the detailed process condition; (5) providing the molecule name that the corresponding molecule properties can be vectorized by their SMILE structure. The data source of collected input data (303 batches $\times$ 44 dimensions of formula factors $\times$ 4 microencapsulation performances) is illustrated in the [Experimental Section](#).

Database standardization guidelines are provided in Table S1, with corresponding theoretical and empirical equations serving as feature selection criteria. As molecular properties are rarely detailed in MIP literature, we employed chemical informatics protocols to investigate their effects within the MIP process. These protocols, illustrated in Figure S1, focus on reaction reactivity and thermodynamic properties. Database features, No.1 through No.29, encompass shell electrophile and hydrogen sources, nucleophile precursors (chain-extender and crosslinker), the target core, cosolvent substances, and the continuous phase solvent. The physical significance and calculation methods for these features are detailed in the Note S1. For shell electrophilic materials, typically prepolymers of common isocyanates and acyl chlorides, we simplified electrophilicity, Hillebrand solubility parameter, enthalpy of formation, and partition coefficient (Log P) calculations. Prepolymer partitioning behavior was estimated from their original molecular structures. Features No.30 through No.44 quantitatively describe the lab-scale MIP operational sequence. Currently, our database excludes Pickering emulsifier-based MIP formulations due to the lack of standard descriptors for nanoscale particle stabilization at the oil-water interface and their jamming behavior [25, 44]. Therefore, we utilize the HLB value, guided by the Bancroft rule, as a sufficient surfactant descriptor for oil-in-water (O/W) and water-in-oil (W/O) emulsions within the MIP process.

The selection of processing parameters is often constrained by data availability in the literature. Critical details such as emulsification times and reagent addition times are rarely reported. In cases of missing data, median values have been imputed to

maintain dataset integrity for algorithmic models. Features such as the dibutyltin dilaurate (DBTDL) concentration, the cosolvent-to-core ratio, and the chain extender-to-crosslinker molar ratio have been thoughtfully included to expand the design space within our materials database. Currently, the scope of shell materials in our repository spans a diverse range of microparticles, including polyurea, polyurethane, and polyamide. Furthermore, the partition coefficient (Log P) for core molecules in our database exhibits a broad range, extending from -2 to 12 , as depicted in Figure 2a,b. The 41 core molecular species cataloged encompass amines, polyols, common solvents, alkanes, esters, ethers, isocyanates, and aromatic derivatives, amongst others.

The multilinearity plot (Figure 2c) of the CPSP relationship of MIP provided a global linear correlation among the features and responses compared with the conventional orthogonal parameter study and design of experiments (DoE) methods with limited material candidates [45–47]. The highlighted green lines of the MIP performances (EE%, μ D and h/R) indicated the general weak linear relationship (Pearson correlation coefficient $R < 0.3$) with 44 input features such as molecule properties and process parameters. As expected, given the diverse physical and chemical properties of MIP formulation elements, our analysis revealed positive or negative cross-correlation among most chemo-physical features. This is particularly evident in the interaction density feature set for dual molecules, calculated from their respective Hillebrand solubilities. However, it is important to note that observed collinearity, such as that between reaction duration and continuous phase solvent properties, does not necessarily imply causality.

The multi-nonlinearity status of the current CPSP relationship reflected the complexity of the MIP process and the difficulty of modeling MIP (Figure 1a). This scenario pushes us to find the causes of nonlinearity and uncertainty. Thus, we found that the possible reasons are: (i) the curated materials diversity; (ii) the errors in data recording and collection; (iii) the combination of hydrophilic (W/O) and hydrophobic (O/W) encapsulation experiments, the hybrid data structure arises from combining hydrophilic (W/O) and hydrophobic (O/W) encapsulation experiments, which exhibit mutually contradictory correlation patterns among the key MIP performance descriptors. In the O/W subset, shell geometry and efficiency are coupled (h/R vs EE%: $r = -0.53$, $p < 10^{-4}$, $n = 266$), whereas size plays essentially no role (μ D vs. h/R : $r = 0.01$, $p = 0.86$; μ D vs EE%: $r = -0.08$, $p = 0.18$). By contrast, in the W/O subset, μ D and h/R become moderately and significantly anticorrelated ($r = -0.40$, $p = 0.02$, $n = 36$), while correlations between h/R and EE% ($r = 0.12$, $p = 0.50$) and between μ D and EE% ($r = 0.20$, $p = 0.24$) are statistically insignificant (Figure 2d).

2.2 | Extracting Microcapsule Synthesis Insights from Interpretable Data-Driven Modeling

Given the independence among MIP performance metrics (Figure 2d), we established three separate regression tasks to unravel the complex property-process-performance relationships. Due to the limited size of the current CPSP database, we carefully selected machine learning algorithms to balance interpretability and accuracy. Our approach incorporated both

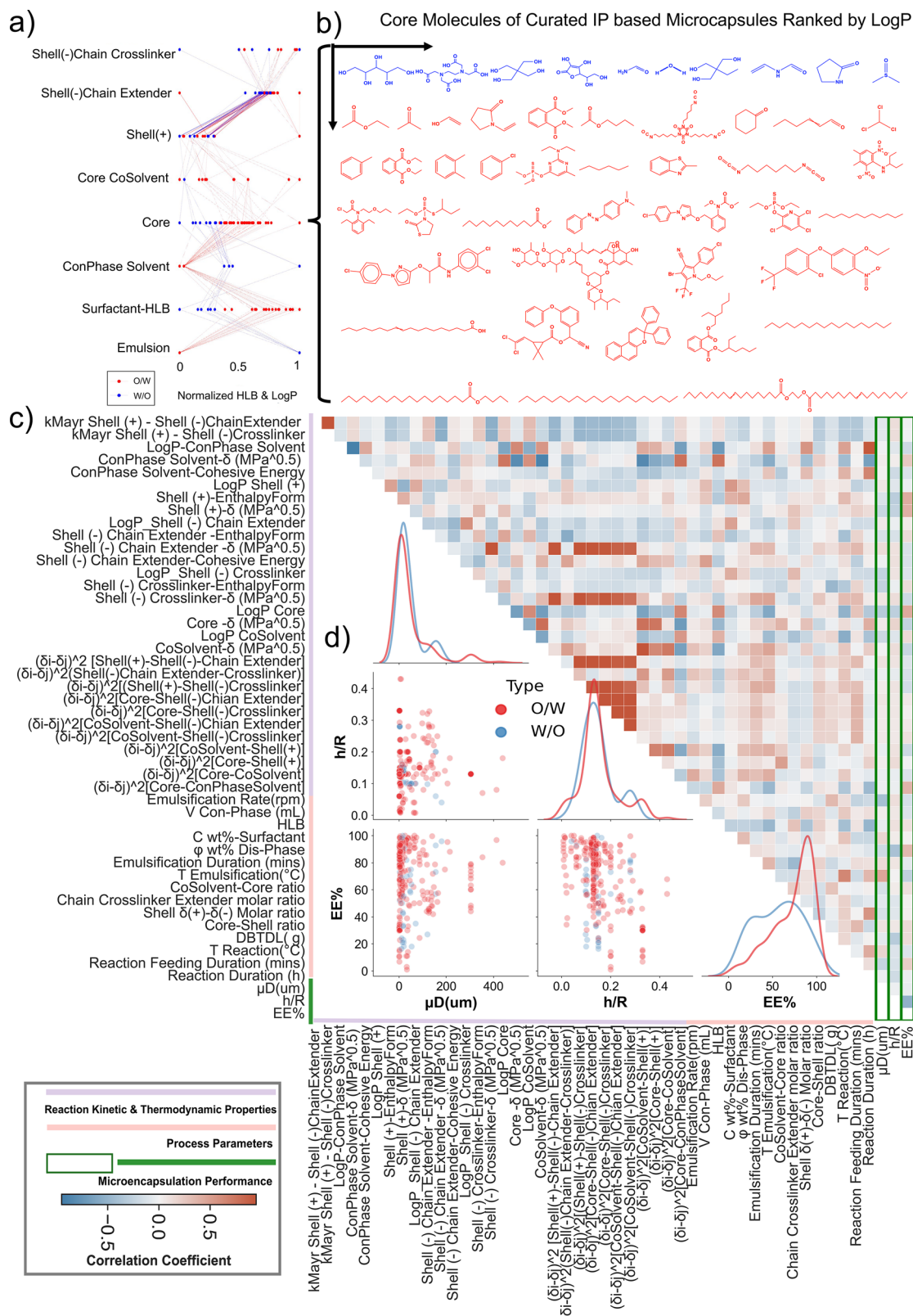


FIGURE 2 | The Chemical-Process-Structure-Performance (CPSP) library of Microencapsulation by Interfacial Polymerization. (a) The hierarchical relationship of existing molecule elements of microcapsules. (b) The diversity of core molecules in the CPSP library. (c) The multicollinearity plot of the property-process-structure-performance linkage of MIP. (d) The independence relationship among the encapsulation efficiency, mean particle diameter, and shell thickness-to-radius ratio.

supervised learning and supervised ensemble methods based on boosting or bagging. This led to the selection of eight regression models, such as LASSO (Least Absolute Shrinkage and Selection Operator), Elastic-Net, Decision Tree, Gaussian-Process-Regression (GPR), Support Vector Regression (SVR), GradientBoost, Random Forest (RF), and XGBoost.

To overcome the overfitting issues, the dataset has already been separated into train (80%) and test (20%) splits. The initial ML pre-training with default hyperparameters has been implemented for comparing the relatively outperforming model by coefficient of determination and root mean square error, as shown in Table S2. Although the XGBoost outperformed in estimating μD and h/R tasks, the main algorithm selection criterion is based on estimation performance on EE%. The EE% estimation is the main consideration in conducting the MIP for designing the functional microparticles. Then, the hyperparameters of the outperformed Random Forest (RF) models for three target performances were tuned by Optuna [48] with a 10-fold grid-search cross-validation method (Table S3 and Figures S2–S4). Under optimal hyperparameter settings, all RF models were optimized for satisfactory estimation performance on the test dataset. Generally, the estimation performances on EE% and h/R are not as robust as the μD task (Figure 3a). The uncertainty of EE% and h/R is mainly from the different characterization methods and the sampling error of measuring the shell thickness by SEM observation.

To delve deeper into the chemical and physical insights of MIP, we employed three interpretative methodologies: the SHAP (Shapley Additive explanations), permutation-based feature importance ranking methods, and the Accumulated Local Effects (ALE) interpretative explanation method [49]. These techniques excel at unraveling complex relationships within datasets. For interpretability analysis of our three performance outcomes, we employed two approaches. First, the SHAP-based feature importance method provides a statistically robust explanation of how physicochemical and procedural variables influence performance predictions. This method considers the distinct physical systems of initial emulsion droplet type (O/W and W/O). Second, the permutation-based feature importance method validates the average prediction effects observed with the SHAP method. Additionally, univariate dependency analysis via the ALE approach reveals the influence of individual features on the associated performance metrics. As depicted in Figure 3b, the top 7 input features that most significantly affect the Random Forest (RF) model's output are listed in descending order of impact. A notable finding, illustrated from left to right in Figure 3b, is that the thermodynamic properties of the Log P Core and $(\sigma_i - \sigma_j)^2$ interaction density (where σ is the Hildebrand solubility parameter) of corresponding compositions are paramount in estimating the overall performance across all three metrics, displaying varied probability density distributions. Subsequent influential features are ordered according to their SHAP and permutation importance, starting with the volume ratio of the disperse phase (φ wt% Dis-Phase), followed by the emulsification rate (rpm), reaction temperature, and duration, and parameters associated with the surfactant, including the HLB and its concentration. The summary of the entire 44 feature importance results is illustrated in Figures S14–S16.

The results from Accumulated Local Effects (ALE) analyses, showcased in Figure 3c and Figures S5–S13, reveal how certain factors correlate with outcomes in three distinct regression tasks. These findings offer insights for the systematic design of experiments, effectively minimizing the dependence on trial-and-error methods in the practical application of microencapsulation by interfacial polymerization. For instance, ALE analysis indicates that a disperse phase volume fraction above 15% tends to yield higher encapsulation efficiencies (EE%) in microcapsules, as evidenced by the upper left quadrant of Figure 3c. Additionally, a sparser interaction density $(\sigma_i - \sigma_j)^2$ between the Shell Electrophile (+) and the Shell Crosslinker is associated with improved EE%, a reflection of increased molecular compatibility. Incorporating a crosslinker into the microparticle shell material produces a more robust and denser barrier compared to using only a chain extender. This reinforced structure effectively impedes core substance migration during both the encapsulation process and long-term storage. Additionally, the formation enthalpy, which is the heat released during molecular assembly, influences the shell material's thermal stability and the resulting membrane's nanoscale structures, such as porosity and roughness. While not currently a standard metric in MIP performance assessments, the nanoscale morphology of the shell membrane still plays a role in influencing the microscale properties, such as the μD and h/R .

Process parameters in MIP, particularly those related to emulsification rate, reaction elements, and surfactant characteristics, are critical in predicting performance outcomes. When estimating the h/R ratio, an emulsification rate of around 1000 rpm is identified as a threshold. A proportional relationship exists between the h/R ratio and emulsification rates from 100 to 1000 rpm. The emulsification rate, indicative of the convection strength of the shear field, depends on the beaker geometry and stirring hydrodynamics. This convection strength, representing hydrodynamic inertia, influences various transport phenomena during the MIP process. These include interactions with dispersed phase viscosity, emulsion droplet interfacial tension, and core molecule diffusion and partition between dispersed and continuous phases. Certain process parameters, particularly those related to chemical reactions and surfactants, significantly impact anticipated performance outcomes. This pronounced effect stems from deliberately selecting these parameters to align with their fundamental roles in the process. Surfactants play a multifaceted role in interfacial polymerization. They not only reduce Gibbs interfacial energy, thereby lowering surface tension, but also alter molecular diffusion rates and partition behavior at the immiscible interface. Moreover, surfactants influence molecular packing density during self-assembly, impacting the organization and structure of the shell at both nano and microscale levels. These effects are crucial as they directly influence the efficacy and integrity of the encapsulated product. The partition coefficient of the core molecule (Log P) exhibits notable differences between oil-in-water (O/W) and water-in-oil (W/O) MIP systems (Figure 3b). Log P is a critical parameter due to its potential influence on various aspects, including core molecule diffusion rate, emulsion droplet stability, molecular compatibility with other substances, and the risk of core substance migration into the continuous phase. As illustrated in Figure 3b and Figure S14a, the interaction density feature $(\sigma_i - \sigma_j)^2$ (Core-Con Phase Solvent) underscores a fundamental formulation design principle: the

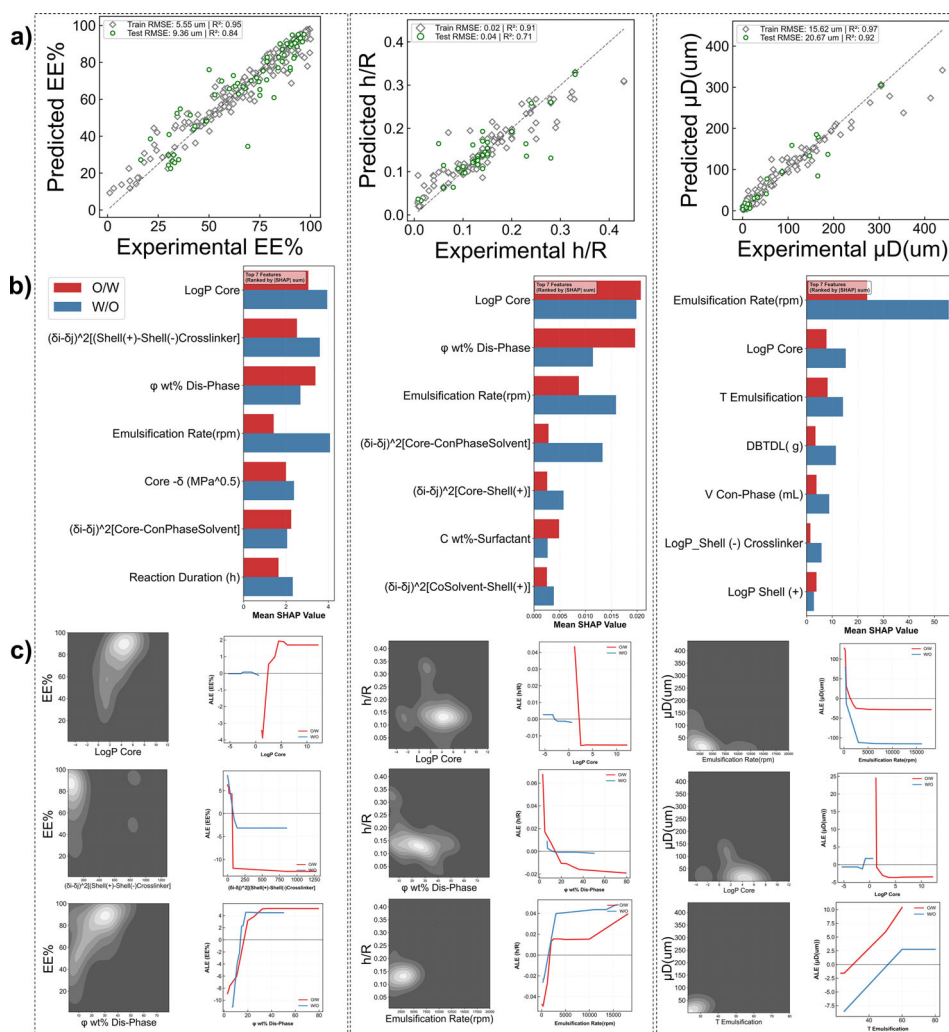


FIGURE 3 | The Performance of Interpretable Machine Learning. (a) The selected and tuned random forest models on predicting encapsulation efficiency, shell thickness, radius ratio, and mean microparticle diameter. The curated dataset is separated into 242 training data points and 60 test data points. The prediction results above are evaluated by root-mean-squared error. (b) The Shapley additive explanations (SHAP) summary analysis for each three random forest models with the top 7 features' importance. The impact of each feature on the corresponding prediction targets is illustrated via decoupling W/O and O/W effects. (c) The top 3 features original dependency (left) and estimation dependency with corresponding targets visualized by the accumulating local effects (ALE) values, which reflect the one-dimensional feature versus estimation target (right).

compatibility between the core and continuous phase solvent is essential for further shell formation and successful encapsulation [50].

2.3 | Extraction of Hidden Microcapsule Rational Design Descriptors Extraction and High Throughput Virtual Screening

Forward estimation is a cornerstone of materials design, yet uncovering hidden knowledge and identifying meaningful design descriptors remains a significant challenge. Accelerating rational design efficiency is particularly critical for MIP. To address this, we developed straightforward yet effective descriptors for rapid virtual screening, designed to handle diverse and often redundant processing conditions. Building on prior section insights, we hypothesize that the average material level encapsulation efficiency ($\mu\text{EE}\% = \frac{\text{EE}\%_1 + \text{EE}\%_2 + \dots + \text{EE}\%_N}{N}$, where N is the batch numbers under the same material composition), under con-

sistent molecule compositions, serves as a robust predictor of expected performance across varying experimental batches with different processing conditions, which denotes a normalized, process-independent encapsulation response.

Insights from the forward machine-learning models (Figure 3; Figures S5–S7 and S14) indicate that encapsulation efficiency is primarily governed by the coupling between thermodynamic interactions and the reactivity of shell-forming species at the emulsion interface. Physically meaningful descriptors are therefore essential for the virtual design of interfacial-polymerization membrane systems before experimental validation.

Within the Damköhler number framework (Equation 1) for interfacial polymerization, the kinetic rate constant of shell formation (k) and reactant concentration (C) define the reaction rate, which competes with diffusive transport characterized by the diffusion coefficient (D) across the interfacial length scale (L^2). Direct in-situ measurement of D at the reaction zone is

difficult because is nanoscopic and diffusion is strongly altered by partitioning between solvent phases with different cohesive energies. To capture this reaction–diffusion competition, we introduce two surrogate descriptors to reveal a hidden relationship between shell-forming behaviors with material-level encapsulation efficiency ($\mu\text{EE}\%$).

The Mayr reactivity parameter provides a quantitative measure of the intrinsic reaction rate of potential shell-forming monomers and is thus highly relevant for interfacial polymerization. In parallel, the partition coefficient $\log P$ is closely related to molecular chemical potential and cohesive energy (Equations S1–S8) and can be interpreted as the Gibbs free-energy change associated with transferring a molecule between immiscible phases (Equation 2). In practical terms, $\log P$ characterizes a molecule's tendency to partition and diffuse across the interface, thereby shaping solvation, local concentration profiles, and the spatial distribution of polymerization sites [11, 18, 19]. Here, $\log P$ values were calculated using RDKit (Wildman–Crippen fragment method), and both the Mayr parameters and $\log P$ are readily accessible from cheminformatics workflows, as detailed in Note S1. These nonlinear descriptors are not only theoretically meaningful but also practically advantageous, allowing for faster and more effective screening of potential molecule combinations.

$$D_a = \frac{\text{Reaction Rate}}{\text{Diffusion Rate}} = \frac{k C L^2}{D} \quad (1)$$

$$\log P = -\frac{\Delta\Delta G_{\text{Transfer}}^{1 \rightarrow 2}}{RT \ln 10} = -\frac{\Delta G^2 - \Delta G^1}{RT \ln 10} \quad (2)$$

Therefore, we identified partition coefficients and Mayr global reactivity parameters as key factors for constructing compositional descriptors in MIP, forming a solid foundation for efficient formula screening (Table 1, top). Variations in material-level encapsulation efficiency ($\mu\text{EE}\%$) are attributed to differences in processing conditions, which triggered microenvironment changes around the oil–water interface, noted as batch effects. To simplify MIP formulation complexity, we categorized the MIP database into four distinct groups based on core composition: (i) core material with a chain extender; (ii) core material with both a chain extender and a crosslinker; (iii) core material with a cosolvent and a chain extender; and (iv) hybrid combinations of these components. This categorization clarifies the interactions within different formulations, streamlining the discovery and optimization of effective microencapsulation strategies.

To efficiently screen a diverse range of designed microparticles, we employed a method combining the symbolic transformer with ordinary least squares (OLS) analysis. We extracted hidden design descriptors using interpretable mathematical expressions. This method generates correlated descriptors that balance complexity and estimation accuracy. The model hyperparameters grid search, symbolic transformer training evolution, and five-fold cross-validation are detailed in Table S4 and Figures S18–S22. To evaluate the collinearity between each descriptor and the mean encapsulation efficiency, we employed the Pearson correlation coefficient, dependency, and response surface as illustrated in Figure S17. The extracted descriptors demonstrate a positive

linear correlation with the $\mu\text{EE}\%$, with correlation coefficients (r) exceeding 0.75. Besides, the models made by the descriptor and OLS analysis have enough screening ability with $R^2 = 0.58$ (Table 1, Bottom). This indicates that the descriptors are descriptive of designing a membrane encapsulation system, providing a feasible tool for guiding the screening of MIP composition (Figure 4b–e).

Furthermore, these descriptors (Table 1, Bottom) can reflect physical meanings in microencapsulation by interfacial polymerization and guiding molecule encapsulation with varying core molecule hydrophobicity, functionality, and expected shell thickness trend.

Based on the Symbolic Transformer results from Table 1, each model reveals the distinct microscopic mechanism governing film formation under specific encapsulation systems. Model A (binary system, baseline encapsulation) captures the fundamental reaction–diffusion competition illustrated in Figure 4a. The descriptor $\left(\frac{-\log P_{\text{SCE}} + \log P_{\text{Core}}^{0.5}}{k_{\text{ECE}} \log P_{\text{SE}}}\right)$ effectively reduces to a $\log P$ Core / k_{ECE} type ratio and quantifies the balance between the diffusive tendency of a hydrophobic core (high $\log P$ Core, O/W) and the interfacial reaction rate of the chain extender. High $\mu\text{EE}\%$ is obtained only when k_{ECE} is sufficiently large to outcompete diffusive loss, consistent with the physical meaning of the interfacial Damköhler number (Equation 1). The partial-dependence map in Figure 4f reveals a narrow optimum: at low k_{ECE} the system is diffusion-dominated [27], whereas overly fast reaction at inappropriate $\log P$ Core shifts the system into a reaction-dominated regime where $\mu\text{EE}\%$ again deteriorates. Practically, Model A defines the design window for conventional microcapsules with single hydrophobic actives and provides a baseline for tuning shell thickness and release stability.

Model B (ternary system, O/W capsules with crosslinker-enhanced shells) highlights a crosslinker-induced “locking” effect but also reveals clear competitive regimes. The descriptor $\left(-\log P_{\text{CPS}} - \log P_{\text{SE}} + \log P_{\text{SCL}} - \log P_{\text{Core}} - \frac{\log P_{\text{Core}}}{\log P_{\text{SE}}}\right)$ encodes the solubility contrast between crosslinker, chain extender, core, and the continuous phase. Moderate positive values drive preferential enrichment of the crosslinker and chain extender at the interface while keeping the O/W core comparatively less soluble in water, which promotes rapid network formation and steric hindrance and thus enhances $\mu\text{EE}\%$. However, the partial-dependence surface in Figure 4g shows a synergy–competition crossover: beyond an intermediate window, excessively high solubility contrast or reactivity leads to a reaction-dominated regime, where overly fast crosslinking and excessive interfacial enrichment create defects or trap the core near the outer shell, reducing $\mu\text{EE}\%$. Owing to its high predictive accuracy ($R^2 = 0.88$), Model B is particularly valuable for functional capsule design where crosslinker chemistry is exploited to tailor shell nano porosity and barrier properties (e.g., for oxidation-sensitive fragrances or controlled-release actives) without sacrificing encapsulation efficiency.

Model C (cosolvent system, miscibility-tuned O/W and O/W like hybrids) reflects miscibility regulation by a secondary solvent. Its descriptor $\left(\frac{\log P_{\text{Core}} \log P_{\text{CoSolvent}}}{3 \log P_{\text{CPS}} + 2 \log P_{\text{Core}}}\right)$ captures how the cosolvent

TABLE 1 | List of selected features with physical meaning explanation (Top) and models for $\mu EE\%$ estimation based on OLS analysis using symbolic transformer extracted hidden descriptors with evaluation matrix, equation boundary, and encapsulation insights (Bottom).

Feature	Description	Physical meaning				Model feature	
$kECE$	The estimated Mayr reactivity coefficient between the shell electrophile and the chain extender	• Estimated reaction rate without considering concentration, steric, and hydrodynamic effects • Facile chemical informatics descriptor				X0	
$kECL$	The estimated Mayr reactivity coefficient between the shell electrophile and the chain crosslinker					X1	
$LogP_{Core}$	The partition coefficient of the core	• Tendency of molecule diffusion across an immiscible interface • Correlating solubility and other thermodynamic behaviors • Facile chemical informatics descriptor				X2	
$LogP_{Cosolvent}$	The partition coefficient of the core cosolvent					X3	
$LogP_{SE}$	The partition coefficient of the shell electrophile					X4	
$LogP_{SCE}$	The partition coefficient of the shell chain extender					X5	
$LogP_{SCL}$	The partition coefficient of the shell chain crosslinker					X6	
$LogP_{CPS}$	The partition coefficient of continues phase solvent					X7	
MIP compositions (Shell electrophile involved)	Models for screening $\mu EE\%$ (y)	R^2	MAE (%)	$LogP_{Core}$	Average h/R	Encapsulation insight	Physical explanation
Core + Chain Extender	$A: y = -0.6 \left(\frac{-LogP_{SCE} + LogP_{Core}^{0.5}}{kECE \ LogP_{SE}} \right) + 85$	0.80	10.9	$-2.31 \sim 7.27$	0.13 ± 0.06	$-W/O$ $-O/W$	Reaction diffusion coordination at the interface without cosolvent solvation and crosslinker

(Continues)

TABLE 1 | (Continued)

Feature	Description	Physical meaning				Model feature
Core + Chain Extender + Chain Crosslinker	$B: y = -4.3(-\text{Log}P_{CPS} - \text{Log}P_{SE} + \text{Log}P_{SCL} - \text{Log}P_{Core} - \frac{\log P_{Core}}{\log P_{SE}}) + 42$	0.88	3.73	3.24	$-0.83 \sim 5.46$	0.09 ± 0.04 -O/W -Multifunction thin shell design Mixing crosslinker and chain extender molecules with relative dense shell for higher EE%
Core + Cosolvent + Chain Extender	$C: y = -12.6(\frac{\text{Log}P_{Core}\text{Log}P_{Cosolvent}}{3\text{Log}P_{CPS}+2\text{Log}P_{Core}})^{0.5} + 102$	0.70	10.76	8.57	$-2.35 \sim 10.15$	0.14 ± 0.08 -W/O -O/W -Thick shell Co-solvent increases miscibility between shell and core molecules
Core + Cosolvent + Chain Extender + Chain Crosslinker	$D: y = -0.11(\frac{\log P_{SCE}^{0.5}}{k_{ECB}^{0.5}k_{ECL}(2\log P_{SE} + \text{Log}P_{Core})^{0.5}}) + 81$	0.58	15.95	13.35	$-3.61 \sim 12.15$	0.12 ± 0.05 -W/O -O/W -Encapsulation discovery -Multifunction design The chain extender molecule has a higher tendency of diffusion compete with the crosslinker, dominating the reaction zone behaviors

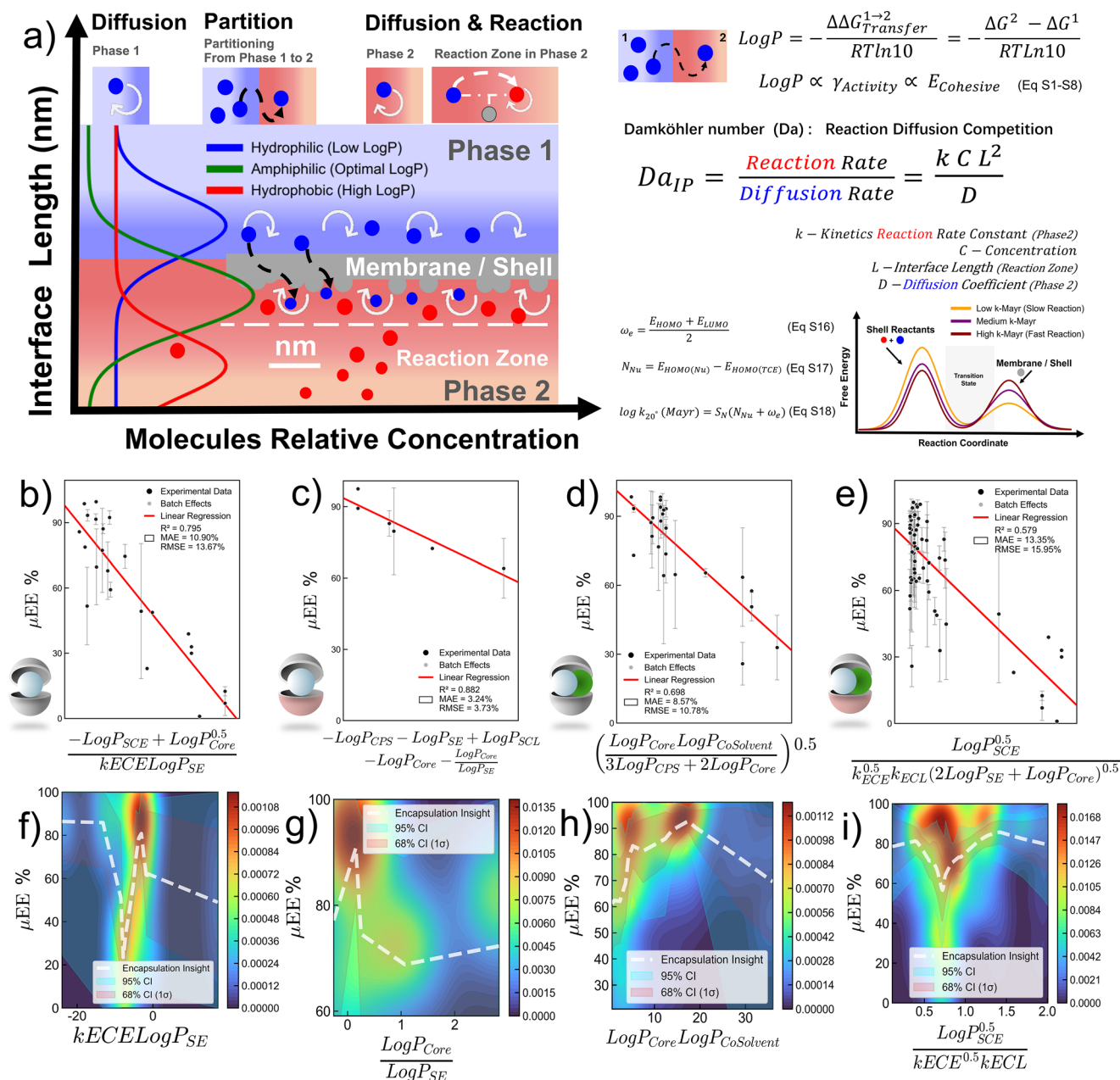


FIGURE 4 | Hidden compositional descriptors for $\mu\text{EE}\%$ and particle morphology extracted by symbolic transformer and partial-dependence analysis. The symbolic transformer was implemented using the Pearson correlation coefficient as the training selection criterion. (a) Schematic illustration of interfacial polymerization, emphasizing the competition between diffusion and reaction at the interface. The process is governed by the partition coefficient ($\log P$), which sets the thermodynamic driving force for phase transfer, and by the Damköhler number for interfacial polymerization (Da_{IP}), which measures the ratio of reaction rate to diffusive transport. The energy diagram (right) shows the activation barriers associated with different reaction kinetics (k -Mayr). (b) Descriptor for a binary system with a single core and a chain extender only (Model A). (c) Descriptor for ternary system with single core, chain extender, and crosslinker (Model B). (d) Descriptor for core/co-solvent mixtures with chain extender only system (Model C). (e) Descriptor for hybrid-molecule formulations (Model D). Error bars represent variability arising from microencapsulation batch conditions. (f–i) Two-dimensional partial-dependence maps showing how pairs of nonlinear descriptor terms control $\mu\text{EE}\%$ in Models A–D, respectively. The white dashed line, warm and cool probability density regions denote regimes of synergistic enhancement and detrimental competition between the underlying physical factors.

modulates the relative solubilities of the core and shell-forming components with respect to the aqueous continuous phase. By reshaping the effective Gibbs freeenergy landscape, the cosolvent adjusts core–shell compatibility and thus filmformation quality and interfacial morphology. The partial-dependence analysis

(Figure 4h) reveals an intermediate “miscibility window” in which $\mu\text{EE}\%$ is maximized; overly good miscibility promotes swelling and leakage, whereas overly poor miscibility leads to interfacial defects and incomplete coverage. This model therefore guides the rational use of cosolvents to formulate more polar

or amphiphilic cores (e.g., polar drugs, reactive monomers) into nominally capsules while retaining high structural integrity.

Model D (hybrid system, high-entropy formulations, and multifunctional capsules) serves as a multicomponent screening tool. The descriptor $\left(\frac{\log p_{SCF}^{0.5}}{kECE^{0.5}kECL(2\log P_{SE} + \log P_{Core})^{0.5}} \right)$ combines the reactivity of chain extender and crosslinker ($kECE$, $kECL$) with their partition behavior relative to the core ($\log P$ Core, typically spanning hydrophobic O/W cores to more amphiphilic payloads). In this most complex formulation space, multiple competing reaction pathways and diffusion processes introduce significant stochasticity; nevertheless, descriptors built solely from $\log P$ and $kMayr$ remain robust (Figure 4e). The partial-bdependence map in Figure 4i shows that the model can still disentangle cooperative regimes, where properly balanced reactivity and partitioning yield high $\mu EE\%$, from detrimental competitive regimes at extreme values. Consequently, Model D enables rapid prescreening of high-entropy interfacial polymerization formulations, supporting the discovery of multifunctional capsules (e.g., multicore, dual-release, or stimulus-responsive systems) without extensive empirical trial-and-error.

These virtual screening descriptors serve as a crucial bridge between material design and practical synthesis. Our findings echo previous observations of a linear correlation between thermal dynamic interaction parameters and encapsulation efficiency by solvent evaporation, suspension polymerization, liposome, and in-situ polymerization encapsulation approaches [31–34]. To validate the effectiveness and correlative capability of our descriptor models and interpretable machine learning frameworks, we implemented a high-throughput virtual screening approach to design virtual microparticles. The workflow for this virtual design process consists of four main steps: (1) selection of core (encapsulant) molecules, (2) identification of shell-forming elements, (3) generation of virtual formulation combinations, and (4) screening of viable formulations. As illustrated in Figure 5a, we began by identifying the target encapsulation objectives, including oil-soluble and water-soluble molecules. The selected core molecules ranged from Beta-carotene (cosolvent: toluene), HMDI, TDI, ethyl octanoate, and limonene, to PVA, DI water, EDTA, TEPA, and PEI (all using DI water as the cosolvent), as shown in the inner circle of Figure 5a. For the shell-forming components, we retrieved nearly 8 million potential chain extenders and crosslinkers from the PubChem database. After applying synthetic constraints such as non-hydrophobicity ($\log P < 1$) and the presence of bifunctional groups, 94 003 candidate molecules were selected, with $\log P$ values ranging from -6 to 1 . In this study, we explored electrophile candidates for shell materials on a limited scale, selecting 2,4-MDI, benzene-1,4-dicarbonyl chloride (TMC), and HMDI as representatives for virtual microcapsule formulation design. As detailed in Table 1, Model B facilitated the simultaneous screening of solvents for the continuous phase.

For oil-in-water (O/W) virtual formulations, water was selected as the continuous phase, while for water-in-oil (W/O) formulations, toluene was used. Once the molecular candidates and constraints were defined, we utilized Model B to generate approximately 40 000 virtual formulations by combinatorially arranging core molecules, chain crosslinkers, electrophiles, and continuous

phase solvents. After filtering out formulations with $\mu EE\%$ below 30% and enforcing criteria for innovative combinations, 1000 virtual formulations were retained for further study. The virtual formulations displayed as a 1D cluster in Figure 5c,d, due to the domain space of the descriptor model being continuous, which reflects the screening ability of hidden descriptors. In addition to general screening, we specifically selected 12 formulations incorporating 2-hydroxyethyl disulfide as the chain extender to enhance shell hydrophobicity and intrinsic self-healing potential. These disulfide-containing microcapsule formulations were prioritized for experimental validation. Furthermore, 17 additional formulations with varying molecular combinations were selected for experimental testing. Our descriptor-based models significantly accelerated microcapsule design using interfacial polymerization. Figure 5b highlights the historical normalized material level design space for interfacial polymerization, encompassing microcapsule formulations collected and synthesized over the past decades under our curation criteria. Leveraging our descriptor framework to rapidly calculate, combine, and screen candidate molecules, the material design space was expanded by 600-fold (as shown in Figure 5c,d). Moreover, even before experimental synthesis, the material level encapsulation efficiency ($\mu EE\%$) range of the resulting microcapsules could be predicted with reasonable accuracy (Figure 5e). These validation experiments demonstrated the robustness and utility of our models in predicting encapsulation outcomes across diverse molecular systems, thereby confirming their generalizability and reliability. While our physically constrained symbolic models extrapolate robustly and yield interpretable descriptors, we did not perform a systematic head-to-head comparison with fully unconstrained models. Unrestricted functional forms may slightly improve apparent accuracy on training and cross-validation sets, but at the cost of more complex, non-physical expressions and a higher risk of overfitting.

In addition to estimating the material level encapsulation efficiency ($\mu EE\%$) of virtual microparticles, their morphological properties play a critical role in formulation screening. In microencapsulation by interfacial polymerization (MIP), two dominant morphologies of microcapsule are observed: “Compact” and “Core–Shell.” These morphologies are primarily governed by the thermodynamic compatibility between the core and the synthesized shell materials, as well as the volume fractions of the individual components [51–53]. However, the role of processing conditions in influencing particle morphology has not been systematically studied or established.

To investigate these morphological characteristics, we standardized and analyzed data from 50 experimental batches, encompassing a wide range of core-shell materials, processing conditions, and resulting morphologies. The dataset is visually represented in a multicollinearity plot (Figure S23). Notably, linear analysis of the dataset revealed no significant linear correlations between particle morphology and the various processing parameters. To develop a classification model for morphology prediction, we selected the partition coefficients of relevant components as input features. Due to the limited size of the morphology-related dataset, we employed decision tree and logistic regression models for classifying morphology tendencies. Among these, the decision tree model outperformed, achieving an impressive classification

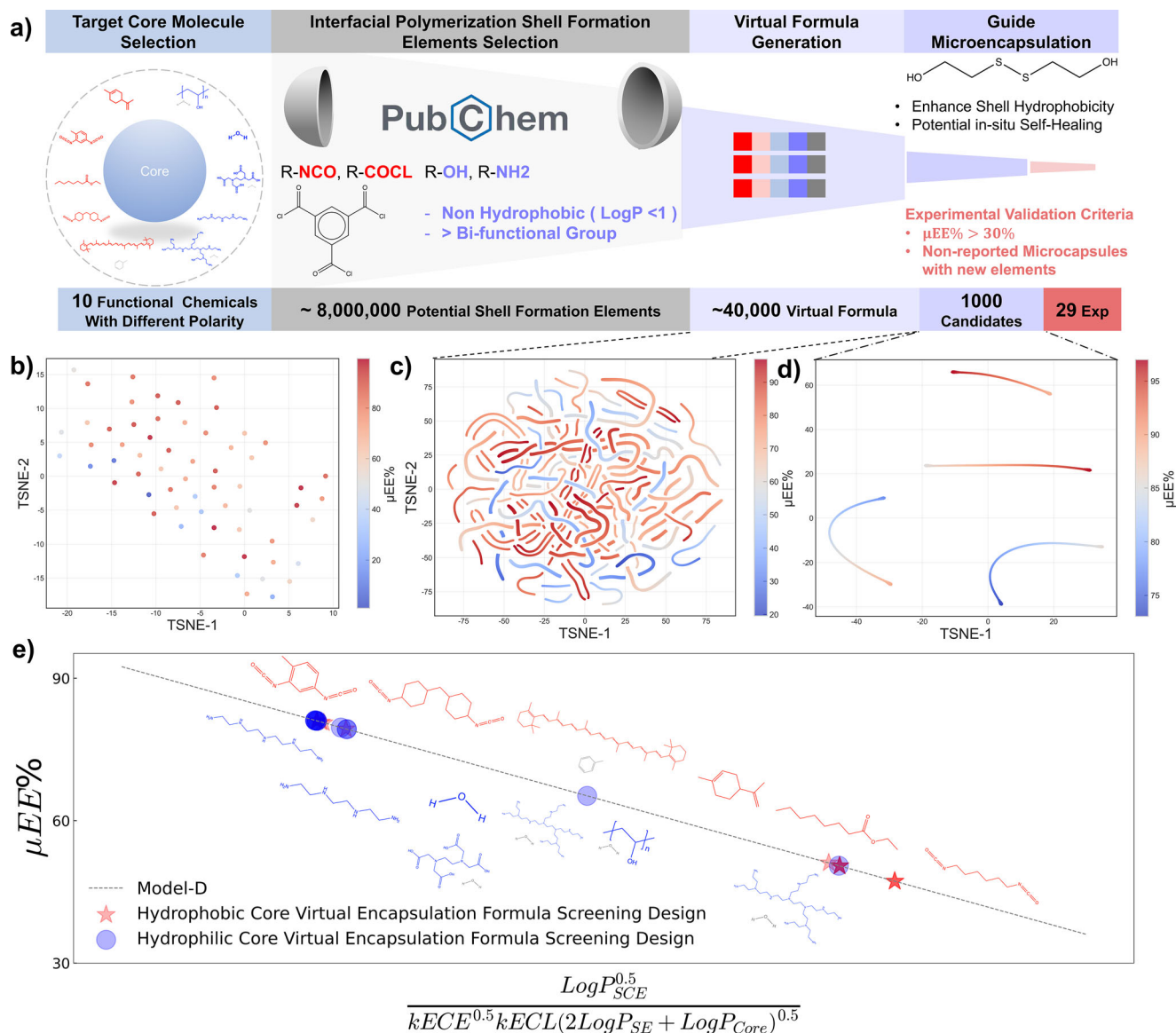


FIGURE 5 | High-Throughput Virtual Design Process for Microencapsulation via Interfacial Polymerization. (a) A schematic representation of the four key steps involved in the virtual design of microcapsules. These steps include the selection of the target core molecule, the identification of appropriate shell-forming components, the generation of virtual formulations, and finally, the experimental synthesis of microcapsules. (b) The dimension reduction visualization of the material design space for the initial library of microcapsules. It does not display individual experimental batches. Rather, it visualizes material-level points obtained by aggregating the batch data for each distinct core-shell molecular combination and mapping them into a normalized descriptor space. For each core-shell system, the molecular descriptors (computed from SMILES) and selected process descriptors are standardized and averaged across all corresponding batches, yielding one representative point per formulation in the chosen descriptor space. (c) The material design space for 40 000 virtual microcapsules, demonstrating the expanded exploration of potential formulations. (d) The refined material design space for 1000 virtual microcapsules, filtered based on the criteria of minimum encapsulation efficiency and the novelty of non-reported microcapsules. (e) The initial molecule encapsulation screening of different designed formulas via extracted compositional descriptors.

accuracy of 0.971 (Figures S24–S26). Feature importance analysis from the decision tree model, specifically for binary classification of morphologies, further confirmed that the nonlinear effects of processing parameters on morphology were also negligible (Figure S25c,d). Instead, molecular properties were identified as the primary factors influencing the final morphology. These findings underscore the effectiveness of the decision tree model in classifying MIP morphologies, making it particularly valuable for predicting and screening morphologies in out-of-sample virtual microparticle designs.

2.4 | The Predictive Applicability of Quantitative Chemical-Process-Structure-Performance Models and Collaborative Guiding of Encapsulating Hydrophobic and Hydrophilic Molecules for Versatile Microcapsules

Recently, Guzik and Allen et al. developed an interpretable machine learning framework to predict the controlled release performance of microparticles produced via the solvent evaporation method of microencapsulation [54]. Their work elucidated

the impact of core and shell molecule properties on release profiles, encompassing both rapid and slow-release rates. This breakthrough has significant implications for accelerating the pharmaceutical applications of microparticles as long injectables. In a complementary study, Ouyang and colleagues leveraged computational techniques to predict drug release profiles and simulate the molecular dynamics of poly (lactic-co-glycolic acid) (PLGA) microsphere formation during solvent evaporation [55]. These studies demonstrate the feasibility of using machine learning to inform experimental microencapsulation.

To validate the performance of previously developed models and follow the insights of high-throughput virtual design, we conducted 29 new validation experiments that were not included in the initial training and testing datasets. These experiments were thoroughly documented and analyzed, with results detailed in supporting tables and figures (Table S5, Figure 6d, Figures S27 and S28 through Figure S40). The validation results, compiled in Table S5, are in fair agreement with the predictions made by three independent random forest machine learning models, as indicated by similar test RMSE values (illustrated by red and blue dots in Figure 6a–c). In addition, the corresponding molecule properties and processing conditions are summarized in Tables S6 and S7. This concordance confirms the robustness of our methods and the chemical-physical insights derived from models.

The 29 validation experiments were divided into two categories based on our analysis: (i). molecule selection for MIP formulas: Experiments #1 to #10 focused on selecting materials for encapsulating hydrophobic and hydrophilic molecules, utilizing specific descriptors and random forest models. (ii). process optimization guided by interpretable machine learning: Experiments #11 to #29 were directed by SHAP value, permutation-based feature importance, and ALE insights to optimize the formula process. In the first category (Experiments #1 to #10), except for #2 and #9 formulas, the shell materials for the microparticles were synthesized using MDI-based polyurethane, varied by different chain extenders and crosslinkers (Figure S27). Notably, Experiment #2 and #9 used TMC as a shell electrophile molecule for encapsulating TDI and TEPA, necessitated by the high reactivity required for high-reactive isocyanate and amine encapsulation, where the average reaction rate of primary amine with aromatic acyl chloride is about several times faster than with aromatic isocyanate. The successful encapsulation of typical molecules was confirmed through FTIR and TGA thermal stability analyses (Figures S29 and S30). Observations of shell microstructure (Figure 6d and Figure S27), μ D (Figure 6c), particle size distribution (Figure S28), thermal behavior of formed shell materials (Figure S31), experimental yields, and loading capacity were also reported in Table S5.

The reduced EE% and weaker diameter-extrapolation performance (R^2) observed for hydrophilic-core microparticles can be attributed to Ostwald ripening. Polar core molecules likely disrupt the stability of the W/O surfactant barrier through hydrogen-bonding interactions, as illustrated in Figure S40. In addition, the hydrophilic-core microparticles exhibit an interaction density with the continuous phase that is, on average, 400 MPa lower than that of oil-core systems, indicating greater mutual solubility with the surrounding medium. Moreover, the

compact morphology of the final microparticles (#3 and #4 in Figure 6d and Figure S34) was consistent with the decision tree model's predictions (Figure S24d) with 52% and 47% porosity. The surface energy profiles of the microparticles were characterized by water contact angle measurements (Figure S32). Notably, microparticles #6 and #20 achieved super hydrophobic status ($\text{WCA}^\circ > 150$), while microparticle #7 demonstrated super hydrophilic properties ($\text{WCA}^\circ < 10$). In addition, the storage performance of the #2 TDI microcapsules was evaluated by TGA after long-term storage. Dried capsules were stored in sealed glass vials at $25^\circ\text{C} \pm 2^\circ\text{C}$ in the dark under sealed conditions (no continuous air exchange) for 120 days. Over this period, the retained core content decreased by approximately 10%. Quantitative image analysis of high-resolution SEM images shows that these capsules possess a nanoscale shell porosity of 13.2%. The time-dependent core loss follows a diffusion-controlled leakage behavior, as confirmed by kinetic fitting (Figure S33i), indicating that the $\sim 10\%$ core loss over 120 days is mainly governed by diffusion through the nano porous shell rather than by shell rupture or chemical degradation. In the second set of experiments (Experiments #11 to #29), we focused on optimizing the encapsulation efficiency (EE%) and microstructure. This was achieved by modifying the shell electrophiles as guided by descriptor screening (Figures S35–S37), adjusting the shell chain extenders and crosslinkers (Figures S33 and S34), and fine-tuning the emulsification rate (TDI and DI Water Core, Figures S33 and S39) and Core shell ratio (TDI Core, Figure S33), the volume fraction of the dispersion phase, the reaction temperature, and the reaction duration for both hydrophobic (HDI core, Figure S38) and hydrophilic encapsulation tasks (DI water core, Figure S39).

Our observations revealed variations in the nanoscale surface morphology of the microparticles, which we attribute to differences in the formation enthalpy of the shell elements and resulting microenvironment instabilities at the emulsion interface. The detailed investigation of microcapsule shell nano porosity and performance linkage will be discussed at next section. While we did not attempt the generative inverse design algorithms [39] for MIP due to its complexities and extrapolation issues, such as high-dimensional parameter spaces and the challenge of discovering unknown mechanism-based microparticle design, our existing interpretable machine learning framework-based high-throughput virtual design still provides significant value. It enables the initial screening of new molecules for encapsulation and supports the synthesis of novel microparticles, thereby expanding the initial MIP performance design space by out-of-sample extrapolation (Figure 6e,f; Table S8). This capability lays a solid foundation for future advancements in this field, enhancing our ability to develop tailored microparticle systems for the following applications.

2.5 | Deciphering the Hidden Role of Nanoscale Structural Descriptors in Governing Microencapsulation Performance and Microcapsules' Function

To elucidate the structure-function nexus, we quantified the surface morphologies of 29 microcapsule variants. Shell nanoporosity emerged as the pivotal bridging parameter. Figure 7a

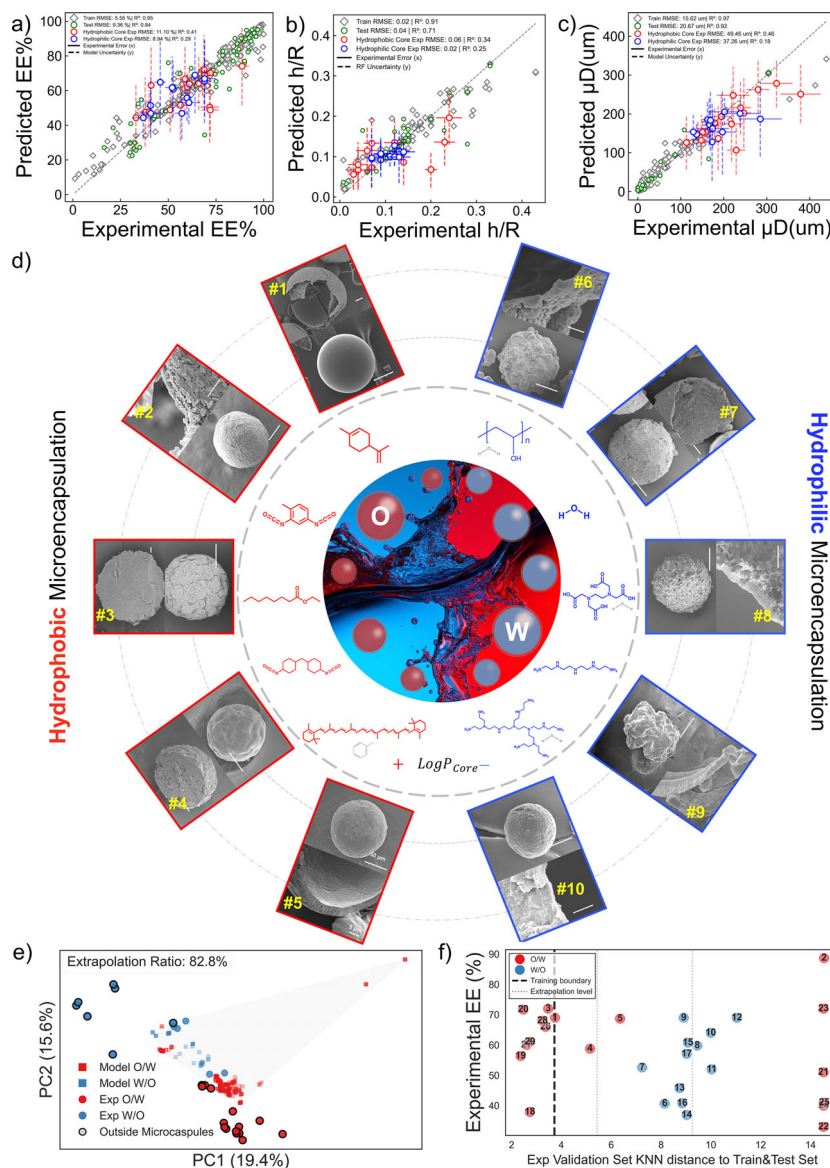


FIGURE 6 | Experimental validation of predicted models for encapsulating a broad range of chemicals from hydrophobicity and hydrophilicity. (a) The extrapolation estimation performance of encapsulation efficiency. (b) The extrapolation estimation performance of the shell thickness radius ratio. (c) The extrapolation estimation performance of the mean particle diameter. (d) The microstructures of validated hydrophobic and hydrophilic core microparticles and corresponding cross-section morphology. (e) Principal Component Analysis (PCA) visualizing the feature space distribution of model training data versus experimental validation points. (f) The microparticle extrapolation domain is defined by using the K-Nearest Neighbors (KNN) distance. Scale bars, 50 μm in (d) inner circle and 5 μm in (d) outside circle. Data are presented as mean values $\pm$ SD ($N = 3$ independent samples).

presents a Pearson correlation (Figure S41) network analysis derived from 29 distinct microcapsule variants. Strikingly, shell nano-porosity emerges not merely as a morphological trait, but as the pivotal “bridging parameter” connecting upstream processing conditions to downstream utility. The network reveals that porosity scales positively with surface roughness (Ra) and microcapsules’ hydrophobicity (WCA $^\circ$), while exhibiting an inverse relationship with shell homogeneity and encapsulation efficiency (EE%). This statistical evidence corroborates an interfacial diffusion-controlled formation mechanism: excessive porosity signifies a destabilized interface where rapid solvent flux disrupts the formation of a compact polymer network, leading to heterogeneous, leaky shells.

The multidimensional dependency of these properties is further mapped in Figure 7b. Porosity is navigated by processing parameters such as surfactant HLB, concentration, and emulsification hydrodynamics (Figure S42); conversely, the WCA $^\circ$ is designed by shell molecule selection and thermodynamic interplay (Figure S43). To rationalize the surface chemistry, we employed Hansen Solubility Parameters (HSP) (Figure 7e). The calculated thermodynamic distances confirm that the shell’s affinity for the dispersion medium dictates its dispersibility, while the introduction of disulfide bonds (–SS–) significantly modulates surface energy. As shown in Figure 7f, the disulfide-functionalized shells exhibit distinct contact angle hysteresis (CAH = $\theta_{Advancing} - \theta_{Retracting}$). This dynamic wettability suggests

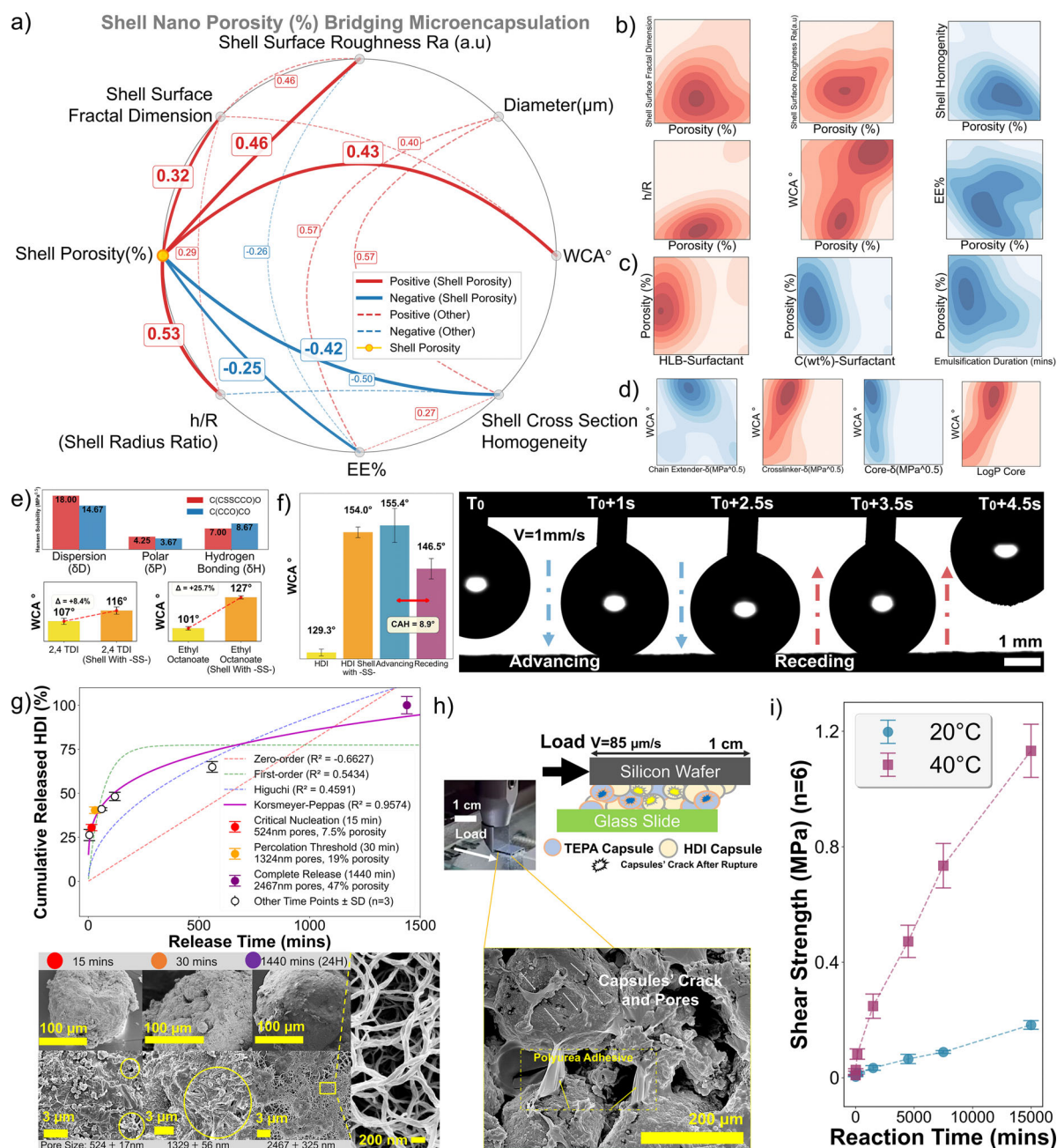


FIGURE 7 | Rational design and multi-scale characterization of porosity-tunable microcapsules for superhydrophobic and self-healing functions. (a) Pearson correlation network analysis reveals shell nano-porosity % as the bridging parameter linking microencapsulation performance, microcapsules' nanostructures, and microstructure to functional properties. Red and blue lines indicate positive and negative correlations, respectively. (b–d) Contour plots mapping the multidimensional parameter space: (b) the influence of porosity on microencapsulation performance, structural descriptors, and water contact angle; (c) the effect of MIP representative processing parameters on controlling porosity; and (d) the chemicals-property relationship governing microcapsule surface wettability. Red and blue contours indicate positive and negative correlations. (e–f) Rational design surface chemistry and wettability analysis: (e) Hansen Solubility Parameters (HSP) explaining the contact angle enhancement by dispersion interaction, comparing #2 and #25, #3 and #26; (f) Impact of disulfide bonds (—SS—) on static and dynamic wettability and Contact Angle Hysteresis (CAH) of HDI microcapsule (#20). (g) Release kinetics and morphological evolution during controlled release: Cumulative release profiles of HDI fitted with various kinetic models (Zero-order, First-order, Higuchi, Korsmeyer-Peppas), correlated with SEM observations of pore and porosity expansion over time. (h–i) Shear strength performance evaluation of adhesive based on HDI(#20) and TEPA(#9) microcapsules: (h) Schematic of the lap-shear test setup and SEM fractography showing the failure cross section of the released polyurea adhesive; (i) Shear strength evolution during the curing process at different temperatures (20°C vs. 40°C), highlighting the heat-accelerated healing efficiency. Data are presented as mean values $\pm$ SD ($N = 3$ independent samples on controlled release characterization, $N = 10$ independent samples on static and dynamic WCA° , and $N = 6$ on shear strength test).

a stimuli-responsive reconfiguration of surface chains, a critical feature for hydrophobic self-cleaning coatings.

The functional significance of this structural control is reflected in the release kinetics of HDI (#20) (Figure 7g and Figure S38(o)). The release profile is best described by the Korsmeyer–Peppas model ($R^2 = 0.9574$), indicating a non-Fickian transport process governed by the interplay between molecular diffusion and macromolecular swelling relaxation. SEM imaging further corroborates this mechanism, revealing a progressive enlargement of nanopores from ≈ 500 nm to ≈ 2.5 μ m, over 24 h. The characteristic chain width of the disulfide-containing polyurethane (–SS–) is ≈ 160 nm (Figure 7g, bottom right), indicating that the linear, dynamic polyurethane segments are preferentially cleaved upon triggering, whereas the more networklike hard domains remain comparatively intact. Together, these observations demonstrate that the porous shell functions as a dynamic “gatekeeper,” responding to internal osmotic pressure rather than acting as a static diffusion barrier. Finally, we translated these microscale mechanisms into macroscopic self-healing behavior using a dual-capsule system (HDI#20 + TEPA#9, Figure 7h). Lapshear tests predominantly exhibit adhesive failure at the adhesive–substrate interface, showing that the healed polyurea layer provides only modest joint strength, although the released species still form a continuous bonding layer across the crack. The adhesive shear strength is temperature-dependent (Figure 7i): curing at 20°C remains diffusion-limited, whereas increasing the temperature to 40°C induces an immediate and pronounced increase in shear strength. For comparison, the neat monomer adhesive tested under identical curing conditions exhibits shear strengths that are, on average, 29% higher than those of the microcapsule system. This Arrhenius-like acceleration highlights the material’s potential for rapid, heat-activated structural repair in demanding engineering applications.

2.6 | Rational Design Guideline for Synthesizing Multi-Functional Microcapsules

Through a systematic data-driven analysis, we established robust quantitative chemical-process-structure-performance relationships for MIP, emphasizing the critical roles of molecular thermodynamics, polymerization kinetics, and process conditions in emulsion-based systems. For initial emulsions with varying polarities, we proposed rational design guidelines, as shown in Figure 8a. Overall, encapsulation efficiency is primarily governed by molecular properties, with precise process control playing a secondary role. Predicting the shell nano and microstructure is more complex [3, 20, 30, 56], as it is significantly influenced by both physicochemical properties and microenvironmental changes under process conditions, warranting further investigation. In terms of particle size distribution, process control has a greater impact, though the molecular physicochemical properties of the core material in the initial emulsion remain an essential consideration for microparticle design. In water-in-oil (W/O) systems, molecular properties and interfacial process control exhibited heightened sensitivity. Due to the thermodynamic incompatibility between hydrophilic cores and polymeric shells, W/O systems predominantly formed core-shell structures, consistent with previous findings [52]. Conversely, in oil-in-water (O/W) systems, molecular design played a more significant

role in determining performance and morphology. The aqueous continuous phase in O/W emulsions, characterized by stable thermodynamic interactions and density differences, facilitated consistent interfacial mass and heat transfer, thereby reducing the impact of process parameters compared to W/O systems. Furthermore, regarding the design principle of shell porosity, the nanostructure is primarily governed by processing conditions, particularly surfactant behavior at the emulsion interface, such as the critical micelle concentration and HLB value. In contrast, the selection of specific molecules and the emulsion type itself play a less critical role. Regarding the water contact angle (WCA°) as a design criterion, the mechanism differs significantly between systems. In an O/W system, the selection of the core molecule critically influences surface wettability. This is because the nanoscale porosity of the shell acts as a capillary network, allowing core chemicals to diffuse outward and modify the surface. Conversely, in microcapsules fabricated via a W/O emulsion, the dependence on the core molecule is less pronounced. While W/O-derived microcapsules can still exhibit hydrophobic behavior, they tend to develop higher shell porosity. This trend is corroborated by our extrapolation validation and functionality data as mentioned in the previous section.

Our analysis identified the key molecular descriptors, such as partition coefficients (Log P) and Mayr reaction reactivities, as critical factors influencing μ EE% across O/W and W/O systems. These descriptors govern molecule diffusion tendency and reactivity at the emulsion interface, offering predictive capabilities for encapsulation performance. Furthermore, these computational descriptors can extend to other microcapsule-based functional systems with varying molecules, energy triggers, and driving forces [58].

Compared to other encapsulation membrane system design protocols (Figure 8b) that rely on diverse descriptors [31, 32, 57], this study uniquely combines molecular screening with processing conditions to achieve improved control over the complex microencapsulation process (Figure 8c). The Ashby plot (Figure 8b) delineates four encapsulation regimes with distinct shell-formation mechanisms. For Solvent Evaporation, this dataset concerns protein encapsulation [31]. EE% reflects a competition between intra-particle cohesion and cross-phase escape during drying. Formulations were guided by Hildebrand solubility parameters, which capture enthalpic core–matrix compatibility but not kinetic or entropic effects such as viscosity build-up or molecule partition. Log P, defined for small solutes, mainly probes compatibility between the liquid core and the protein phase rather than the protein itself. Because protein mobility is strongly constrained, EE% is dominated by aggregation and film-formation dynamics, and EE% shows almost no correlation with Log P ($R^2 \approx 0$). Improving EE%, therefore, requires combining cohesive/partition design with strict process control.

Suspension Polymerization was designed and analyzed using Hansen solubility parameters, which capture dominant hydrogen-bonding interactions and predict EE% accurately [32]. In contrast, Log P merges dispersion, polar, and hydrogen-bonding contributions into a single scalar and cannot resolve the specific hydrogen bonding governing system, highlighting a key

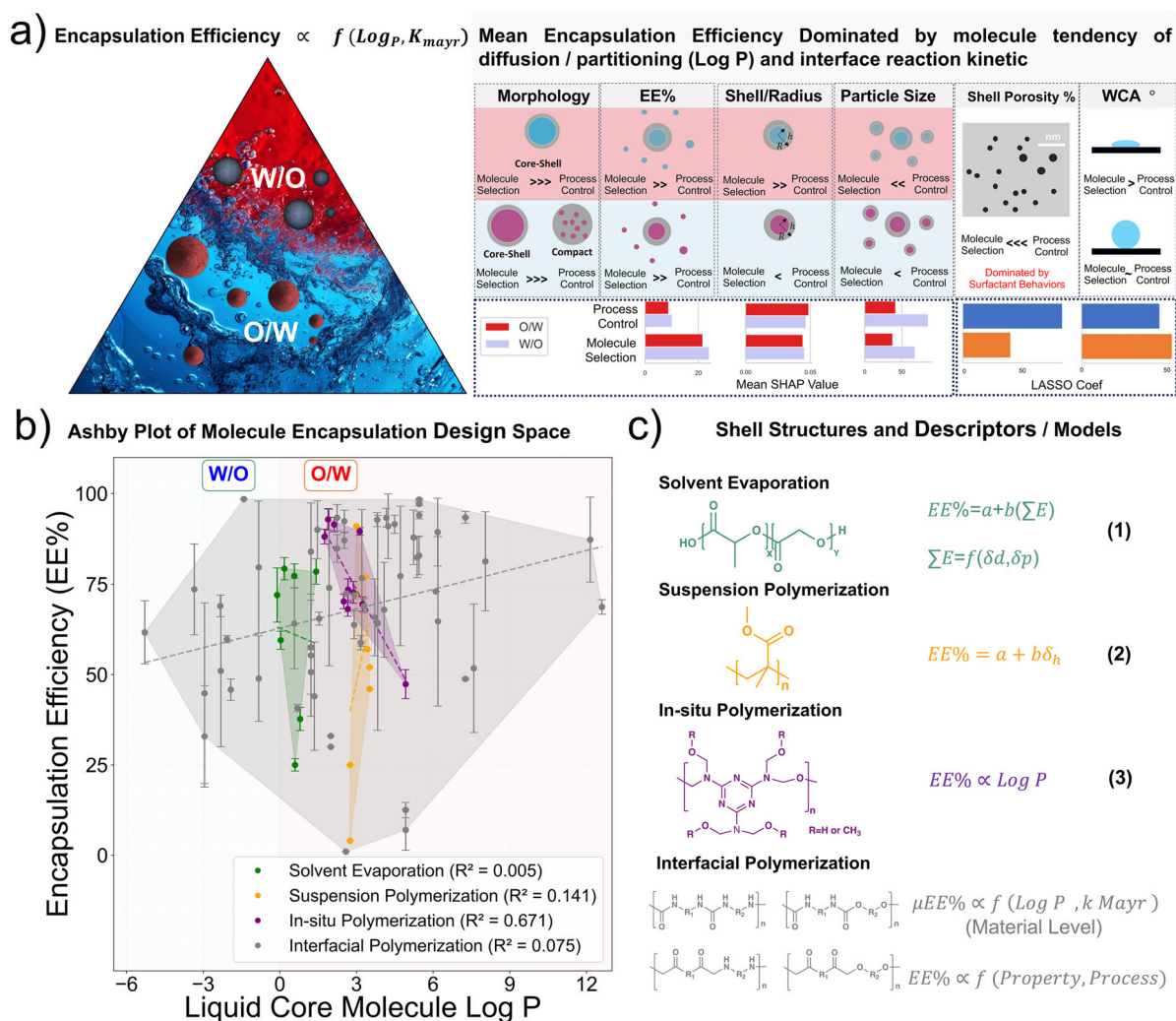


FIGURE 8 | Rational Design Guideline for Microencapsulation by Interfacial Polymerization and Comparison of existing encapsulation membrane system design tools for predicting encapsulation efficiency (EE%). (a) the molecule selection and process control trade off on different MIP performance and emulsion systems. (b) The Ashby plot of encapsulation system selection illustrated with Log P of core molecules; (c) The shell structures of different microencapsulation protocols and the comparison of corresponding descriptors/models for predicting EE%, c1 is from [31], c2 is from [32], and c3 is from [57].

limitation of Log P for interaction-dominated formulations. Log P remains useful for rapid molecular screening and mapping the broader design space, whereas Hansen descriptors are better suited for detailed optimization within a given chemistry. In situ Polymerization produces dense, low-porosity shells. EE% correlates moderately with core hydrophobicity ($R^2 \approx 0.67$), indicating that partition into the compact wall is a primary determinant [57]. These systems strongly protect volatile or reactive cores but often rely on chemistries with environmental and biocompatibility concerns, and generally do not permit one-step encapsulation of fully aqueous cores.

Interfacial Polymerization spans the broadest Log P–EE% design space yet is least amenable to linear prediction: EE% versus Log P shows only a weak correlation ($R^2 \approx 0.08$) because shell structure emerges from a coupled balance of diffusion, partitioning, and interfacial reaction kinetics. Our reaction–diffusion–partition framework, combining Log P with kMayr-based reactivity, overcomes this limitation: for interfacial-polymerization data, symbolic models A–D (Table 1 and Figure 4b–i) increase the predictive

R^2 for material level encapsulation efficiency ($\mu EE\%$) from ≈ 0.08 to 0.58–0.88, reducing error by about an order of magnitude and enabling efficient screening.

Across the Log P–EE% landscape, these mechanistic constraints translate into distinct design spaces. Interfacial polymerization can directly encapsulate water-soluble molecules from aqueous phases, accessing a wide low- and mid-Log P region. Suspension polymerization can also process W/O emulsions with dispersed aqueous actives. By contrast, solvent evaporation typically requires multiple emulsions (e.g., W/O/W) to host hydrophilic cores, and in situ polymerization generally cannot encapsulate purely aqueous cores in a single step. Because many drugs carry hydroxyl and amino groups and hence fall into this hydrophilic regime, interfacial polymerization emerges as a particularly convenient and versatile platform for drug encapsulation, if shell chemistry is tuned for biocompatibility. However, in industrial applications involving complex mixtures or encapsulation of unknown core materials, single molecular property descriptors may fail. In such cases, encapsulation efficiency must be

re-modeled and predicted using more intricate measurements, such as interfacial tension and viscosity [34]. Nonetheless, this Log P-EE% Ashby plot highlights MIP's potential to encapsulate a wide range of matters, including proteins and peptides with different polarities. Additionally, MIP enables the design of functional, biocompatible polyurethane and polyurea shell materials, demonstrating its capability to tackle more complex bioencapsulation challenges.

Using interpretable machine learning, we identified nonlinear relationships between molecular descriptors, processing parameters, and performance metrics, showcasing the versatility of MIP as a platform for tailoring microstructures (Figures 6d and 8c). We systematically integrated Log P values and Mayr reactivity parameters for microparticle design, establishing predictive descriptors and comprehensive machine-learning models. Although the extracted insight from these descriptors (Table 1) has room for high accuracy, limiting our findings to correlation identification. Accordingly, the primary objective of these descriptors is not to maximize numerical prediction accuracy, but to establish a physically grounded, interpretable design framework. We deliberately trade a small amount of potential predictive performance for mechanistic transparency and generalizability that can support reliable extrapolation and rational microcapsule design. Validation and theoretical derivation of these encapsulation efficiency correlations are beyond this study's scope. Additionally, exploring the interaction between interfacial enthalpy and Gibbs' membrane elastic energy during MIP presents a promising research direction. Overcoming challenges in morphological control will require multiscale simulations combining *ab initio* molecular dynamics and phase-field methods. Specifically, we will focus on predicting the shell surface morphology patterns that arise from polymerization-induced phase separation [59] and exploring the nano- and micro-scale Turing patterns that emerge by modifying the diffusion coefficients of shell formation elements [10]. Furthermore, the evolution of shell microstructure and nanoscale morphology, shaped by shell formation dynamics, calls for rigorous theoretical studies to address these complexities.

3 | Conclusion

This study transcends the conventional application-specific boundaries of interfacial polymerization (IP) by introducing a data-driven platform that programs IP outcomes through interpretable machine learning. We move beyond the empirical paradigm that has long constrained not only microencapsulation (MIP) but the broader IP field, by deciphering mechanisms oriented quantitative design rules exemplified by key descriptors such as monomer partition coefficient. Our framework establishes a predictive, quantitative chemical-process-structure-performance relationship, enabling the programmable design of polymeric microcapsules with tailored encapsulation efficiency, size, and shell architecture. The success in encapsulating diverse payloads demonstrates the robustness of this platform. Crucially, the insights we extract are not limited to microencapsulation. The fundamental descriptors and design principles unveiled here provide a systematic framework for rational IP design.

This work marks a paradigm shift from trial-and-error to a predictive and programmable approach in materials synthesis. The implications are immediate for microencapsulation and extend broadly to the rational development of advanced polymer films, catalytic microreactors, digital cells, and next-generation functional membranes. By providing a foundational platform that bridges molecular-level understanding to macroscopic performances, this research opens new avenues for accelerating innovation across the entire landscape of interfacial polymerization.

4 | Experimental Section

4.1 | Materials

The 4,4-Diphenylmethane diisocyanate (MDI) prepolymer, Suprasec 2644, was kindly provided by Huntsman Corporation. Additional chemicals, including Tetraethylenepentamine (TEPA), Triethylenetetramine (TETA), Diethylenetriamine (DETA), 4,4'-Methylenebis(cyclohexyl isocyanate) (HMDI, a mixture of isomers), Toluene-2,4-diisocyanate (TDI), Hexamethylene diisocyanate (HDI), Gum Arabic (GA, an oil-in-water surfactant), Polyvinyl Alcohol (PVA-1788, serving as both O/W surfactant and W/O core material), 1,4-Butanediol (BDO), Ethylenediaminetetraacetic acid (EDTA), Polyethyleneimine (PEI, with a molecular weight of approximately 13,000, in a 50wt% aqueous solution), Thiodiglycol, n-hexadecane and Toluene were procured from Sigma-Aldrich. Furthermore, Benzene-1,4-dicarbonyl chloride (TMC), 2-Hydroxyethyl disulfide, Limonene, Ethyl octanoate, Dibutyltin dilaurate (DBTDL), Arlacel P135 (an O/W surfactant), Beta-carotene, and Hexane were acquired from Aladin. Co. Limited. All these chemical reagents were utilized as received, with no additional purification.

4.2 | Microcapsules Synthesis

The general preparation methodology for microencapsulation via interfacial polymerization consists of several sequential and well-defined stages, like those described in previous works [60, 61]: i) droplet generation: The continuous phase is first prepared by mechanically stirring a phase solvent and a chosen surfactant. Next, a dispersion phase is created by combining either electrophilic polymers (for hydrophobic encapsulation) or nucleophilic polymers (for hydrophilic encapsulation) with a core monomer and a cosolvent. This dispersion is then emulsified into the continuous phase under mechanical agitation, with temperature controlled via a water bath. ii) shell generation: Shell-forming molecules (either nucleophilic for hydrophobic encapsulation or electrophilic for hydrophilic encapsulation) are introduced in a controlled manner. This is achieved through precise dropwise addition, alongside specific heating protocols. Reaction conditions are maintained for predetermined intervals to ensure the formation of the targeted microparticles. iii) cooling, washing, and drying: Post-formation, the microparticles are cooled and washed with solvents of varying polarities (distinct from the encapsulated core molecules) to remove resid-

ual components. Finally, the microparticles are collected for further analysis. Detailed standardized formulation factors and process conditions of extrapolation validation are provided in Tables S1, S6, and S7.

4.3 | Microcapsules Characterization

Thermogravimetric analysis (TGA) was conducted using a TA Q5000 thermal analyzer in a nitrogen atmosphere, with a temperature range from 25°C to 500°C and a heating rate of 15°C/min. For the stability of these tests, samples including microparticles, shell powders, or liquid cores, weighing between 2.0 and 3.0 mg, were used. The TGA data facilitated the determination of core fractions within the microcapsules. Using these characterized core fractions, encapsulation efficiency (EE%) and loading capacity were calculated as Equations (3) and (4). Equation (5), which involves the measurement of the final product mass, was used to calculate the batch-based yield.

$$\text{Encapsulation Efficiency (\%)} = \frac{\text{Measured Core Content \%}}{\text{Designed Core Content \%}} \quad (3)$$

$$\text{Loading Capacity (\%)} = \frac{\text{Mass of Core Extracted}}{\text{Mass of Final Products}} \quad (4)$$

$$\text{Yield (\%)} = \frac{\text{Mass of Final Products}}{\text{Mass of Total Input Components}} \quad (5)$$

The morphologies of the microparticles were examined using field emission scanning electron microscopy (FESEM, JEOL-6700). The mean diameter of the microcapsules and their size distribution were quantified through analysis of SEM images, with at least 120 measurements being processed using ImageJ software. Water contact angle measurements were performed on microparticle samples prepared in a powder state and adhered with epoxy resin. The water contact angle was observed and measured at least 10 times per sample using Theta Flow equipment. Core encapsulation verification was conducted using Fourier-transform infrared (FTIR) spectroscopy with a Bruker Vertex 70 spectrometer. Spectra ranging from 400 to 4000 cm⁻¹ were recorded to analyze the chemical composition of the encapsulated cores. The glass transition temperature of the synthesized shell materials was evaluated using Differential Scanning Calorimetry (DSC). For these measurements, fragments of the shell wall, obtained from crushed and dried particles, were heated from 25°C to 240°C at a rate of 15°C/min, also in a nitrogen atmosphere. For controlled-release experiments, the UV-vis spectra of HDI released into hexane were recorded at different time points using a PerkinElmer Lambda UV/Vis spectrophotometer over the wavelength range of 200–500 nm (Figure S38o). The adhesive shear strength was measured using a DAGE 1000 bond tester, with the test configuration shown in Figure 7h. The lower test chip was rigidly fixed to the substrate, while a flat shear tool was applied to the upper chip. In accordance with the JEDEC JEP167A protocol, a constant displacement rate of 85 μm s⁻¹ was imposed until bond failure occurred.

4.4 | Dataset Preparation and Mechanisms-Based Feature Standardization

We curated a batch-level dataset of 353 experimental microencapsulation batches performed via in situ interfacial polymerization, recent literature reports [45–47, 60–96]. These batches cover 110 distinct core-shell microparticle systems. For each batch, we extracted the reported processing conditions together with three key physical responses: encapsulation efficiency (EE%), mean capsule diameter, and shell-thickness-to-radius ratio. Most of these batches are literature-derived, with a smaller subset generated in-house. In this study, we analyzed 302 microcapsule batches with complete experimental records (44 formulation variables, four performance metrics, emulsion type, and surfactant identity). Microparticle morphology (compact vs. core-shell) was obtained from an external literature dataset of 51 formulations [52, 53], which served solely as a morphology-classification model source and was not included in the quantitative EE% correlations because performance data were not available.

Published microencapsulation studies cover diverse application domains and often report incomplete or heterogeneous information; therefore, not all available reports are suitable for systematic modelling. To obtain an internally consistent, ML-ready dataset, we applied explicit inclusion criteria when selecting literature sources. Studies were considered for inclusion if they reported at least (i) the core mass or volume fraction for different experimental batches, (ii) the encapsulation efficiency (EE%) for those batches, and (iii) the detailed process conditions (e.g., emulsification method, stirring speed, temperature profile, reaction time). Whenever available, we additionally required (iv) structural information on the capsules, such as particle-size distribution and/or cross-sectional morphology, and (v) unambiguous identification of the core and shell molecules that could be converted into molecular structures (IUPAC or common names convertible to canonical SMILES). In practice, only studies satisfying criteria (i)–(iii) and providing at least some information relevant to (iv) or (v) were retained in the final dataset. Literature sources were initially identified by keyword queries in major databases using combinations of “microencapsulation by interfacial polymerization”, “polyamide microcapsule/microsphere,” “polyester microcapsule/microsphere,” “polyurethane microcapsule/microsphere,” and “polyurea microcapsule/microsphere,” followed by manual screening against the above inclusion criteria.

For each core and shell molecule, canonical SMILES strings were compiled from the chemical names reported in the literature and in our in-house records and were used to compute molecular descriptors (log P, k-Mayr), where the detailed computation approaches are described in detail in the Note S1. These molecular descriptors, together with selected process descriptors (core fraction, emulsification agitation conditions, temperature, etc.), were standardized and used as inputs to the machine-learning models. The feature construction and standardization procedures are described in detail in the Supporting Information (Note S1 and Table S1). In addition, a curated subset of 100 batches of microencapsulation data is released as the representative dataset, including component SMILES, process conditions, performance

metrics, emulsion type, surfactant name, and DOI (Tables S9 and S10) [45, 46, 68, 70, 88, 89, 96].

4.5 | Machine Learning Model Development and Interpretation

In total, twenty-four regression models were trained and investigated for the corresponding three tasks. These included LASSO, Elastic-Net, Decision Tree, Gaussian-Process-Regression (GPR), Support Vector Regression (SVR), GradientBoost, Random Forest (RF), and XGBoost models. Two classification models were trained for the morphology propensity screening task, such as logistic regression and a decision tree. All these models were implemented in Python. Except that the XGBoost models were built using the XGBoost package [97], all other models, including the K Nearest Neighbors (KNN), Principal Component Analysis (PCA), and T-Distributed stochastic neighbor embedding (T-SNE), were built by the Scikit Learn library [98]. SHAP, permutation feature importance, and ALE analysis were conducted on the three trained 44-feature RF models. The effect of the various input features on the encapsulation efficiency, h/R, and μ D predictions for the initial dataset was assessed using the TreeSHAP package, Scikit Learn, and the ALE dependency plot visualizations from the SHAP and ALE library in Python.

4.6 | Hidden Descriptor Extraction and Surface Structural Descriptor Calculation

Symbolic transformer (ST) is a numerical induction machine learning model with an explicit mathematical expression by primary math operators rather than fixed prior statistical assumptions. Symbolic transformer (ST) was implemented by gplearn [99]. In this work, a symbolic transformer is used to help extract straightforward and effective hidden descriptors and link with Ordinary Least Squares to form the MIP formula screening models. Shell nanoscale porosity was quantified by extracting shell surface roughness, shell homogeneity, and shell surface fractal dimension from SEM images of the microcapsule shell surfaces and cross-sections using ImageJ.

Author Contributions

J.Y. conceived the idea and supervised the project. Y.W. and B.T. provided resources. Y.H. led dataset construction and analysis. Y.H. and C.Q. carried out computational studies and analysis. Y.H., W.D., Y.Z., M.L. and Y.Z. performed experimental synthesis and characterization. J.Y. and Y.H. wrote the paper.

Acknowledgements

This work was supported by the Hong Kong Research Grant Council (RGC) via Strategic Topics Grant (Grant #: STG2/E-605/23N), the Mainland-Hong Kong Joint Funding Scheme (Grant #: MHP/005/22), the Innovation and Technology Commission of Hong Kong (ITC-CNERC14SC01), and the Department of Science and Technology of Guangdong Province (Grant #: 2024A0505040011). The authors are grateful to the Materials Characterization and Preparation Facility (MCPF) of HKUST for their assistance in experimental characterizations. We thank Dr. Yifan CAO for insightful discussions on visual design and Dr. Zunyu GUAN for assistance with characterization.

Conflicts of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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